

81. $\tan^{-1}\left(\frac{1+\sqrt{3}}{3+\sqrt{3}}\right) + \sec^{-1}\left(\sqrt{\frac{8+4\sqrt{3}}{6+3\sqrt{3}}}\right)$ is equal to :

$\tan^{-1}\left(\frac{1+\sqrt{3}}{3+\sqrt{3}}\right) + \sec^{-1}\left(\sqrt{\frac{8+4\sqrt{3}}{6+3\sqrt{3}}}\right)$ बराबर है :

(A) $\frac{\pi}{6}$

(B) $\frac{\pi}{2}$

(C) $\frac{\pi}{4}$

(D) $\frac{\pi}{3}$

Ans. D

Sol. $E = \tan^{-1}\left(\frac{1+\sqrt{3}}{\sqrt{3}(\sqrt{3}+1)}\right) + \sec^{-1}\left(\frac{16+8\sqrt{3}}{12+6\sqrt{3}}\right)^{1/2}$

$$\therefore \frac{16+8\sqrt{3}}{12+6\sqrt{3}} = \frac{4(\sqrt{3}+1)^2}{(3+\sqrt{3})^2}$$

$$\therefore \left(\frac{16+8\sqrt{3}}{12+6\sqrt{3}}\right)^{1/2} = \frac{2(\sqrt{3}+1)}{3+\sqrt{3}} = \frac{2}{\sqrt{3}}$$

$$E = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) + \sec^{-1}\left(\frac{2}{\sqrt{3}}\right) = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) + \cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$$

$$E = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) + \cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$$

$$= \frac{\pi}{6} + \frac{\pi}{6} = \frac{\pi}{3}$$

82. If $\sin^{-1}\frac{\alpha}{17} + \cos^{-1}\frac{4}{5} - \tan^{-1}\frac{77}{36}$, $0 < \alpha < 13$ then $\sin^{-1}(\sin \alpha) + \cos^{-1}(\cos \alpha)$ is equal to

यदि $\sin^{-1}\frac{\alpha}{17} + \cos^{-1}\frac{4}{5} - \tan^{-1}\frac{77}{36}$, $0 < \alpha < 13$ है, तो $\sin^{-1}(\sin \alpha) + \cos^{-1}(\cos \alpha)$ बराबर है :

(A) 16

(B) 0

(C) π

(D) $16 - 5\pi$

Ans. C

Sol. $\cos^{-1}\frac{4}{5} = \tan^{-1}\frac{3}{4}$

$$\therefore \sin^{-1} \frac{\alpha}{17} = \tan^{-1} \frac{77}{36} - \tan^{-1} \frac{3}{4} = \tan^{-1} \left(\frac{\frac{77}{36} - \frac{3}{4}}{1 + \frac{77}{36} \cdot \frac{3}{4}} \right)$$

$$\sin^{-1} \frac{\alpha}{17} = \tan^{-1} \frac{8}{15} = \sin^{-1} \frac{8}{17}$$

$$\Rightarrow \frac{\alpha}{17} = \frac{8}{17} \Rightarrow \alpha = 8$$

$$\begin{aligned} \therefore \sin^{-1}(\sin 8) + \cos^{-1}(\cos 8) \\ = 3\pi - 8 + 8 - 2\pi \\ = \pi \end{aligned}$$

83. Let $a_1 = 1, a_2, a_3, a_4, \dots$ be consecutive natural numbers.

Then $\tan^{-1} \left(\frac{1}{1+a_1 a_2} \right) + \tan^{-1} \left(\frac{1}{1+a_2 a_3} \right) + \dots + \tan^{-1} \left(\frac{1}{1+a_{2021} a_{2022}} \right)$ is equal to

माना $a_1 = 1, a_2, a_3, a_4, \dots$ क्रमागत धन पूर्णांक है। तब

$\tan^{-1} \left(\frac{1}{1+a_1 a_2} \right) + \tan^{-1} \left(\frac{1}{1+a_2 a_3} \right) + \dots + \tan^{-1} \left(\frac{1}{1+a_{2021} a_{2022}} \right)$ बराबर है :

(A) $\tan^{-1}(2022) - \frac{\pi}{4}$ (B) $\frac{\pi}{4} - \cot^{-1}(2022)$ (C) $\frac{\pi}{4} - \tan^{-1}(2022)$ (D) $\cot^{-1}(2022) - \frac{\pi}{4}$

Ans. A,B

Sol. $a_2 - a_1 = a_3 - a_2 = \dots = a_{2022} - a_{2021} = 1.$

$$\therefore \tan^{-1} \left(\frac{a_2 - a_1}{1 + a_1 a_2} \right) + \tan^{-1} \left(\frac{a_3 - a_2}{1 + a_2 a_3} \right) + \dots + \tan^{-1} \left(\frac{a_{2022} - a_{2021}}{1 + a_{2021} a_{2022}} \right)$$

$$\begin{aligned} & \left[\left(\tan^{-1} a_2 \right) - \tan^{-1} a_1 \right] + \left[\tan^{-1} a_3 - \tan^{-1} a_2 \right] + \dots + \left[\tan^{-1} a_{2022} - \tan^{-1} a_{2021} \right] \\ & = \tan^{-1} a_{2022} - \tan^{-1} a_1 \end{aligned}$$

$$= \tan^{-1}(2022) - \tan^{-1} 1 = \tan^{-1} 2022 - \frac{\pi}{4} \text{ (Option 3)}$$

$$= \left(\frac{\pi}{2} - \cot^{-1}(2022) \right) - \frac{\pi}{4}$$

$$= \frac{\pi}{4} - \cot^{-1}(2022) \text{ (Option 1)}$$

84. Let $(a, b) \subset (0, 2\pi)$ be the largest interval for which $\sin^{-1}(\sin \theta) - \cos^{-1}(\sin \theta) > 0$, $\theta \in (0, 2\pi)$, holds. If $\alpha x^2 + \beta x + \sin^{-1}(x^2 - 6x + 10) + \cos^{-1}(x^2 - 6x + 10) = 0$ and $\alpha - \beta = b - a$, then α is equal to :

(A) $\frac{\pi}{48}$

(B) $\frac{\pi}{16}$

(C) $\frac{\pi}{8}$

(D) $\frac{\pi}{12}$

Ans. D

Sol. $\sin^{-1} \sin \theta - \left(\frac{\pi}{2} - \sin^{-1} \sin \theta \right) > 0$

$$\Rightarrow \sin^{-1} \sin \theta > \frac{\pi}{4}$$

$$\Rightarrow \sin \theta > \frac{1}{\sqrt{2}}$$

$$\text{So, } \theta \in \left(\frac{\pi}{4}, \frac{3\pi}{4} \right)$$

$$\theta \in \left(\frac{\pi}{4}, \frac{3\pi}{4} \right) = (a, b)$$

$$\Rightarrow b - a = \frac{\pi}{2} = \alpha - \beta$$

$$\Rightarrow \beta = \alpha - \frac{\pi}{2}$$

$$\Rightarrow \alpha x^2 + \beta x + \sin^{-1}[(x-3)^2 + 1] + \cos^{-1}[(x-3)^2 + 1] = 0$$

$$x = 3, 9\alpha + 3\beta + \frac{\pi}{2} + 0 = 0$$

$$\Rightarrow 9\alpha + 3\left(\alpha - \frac{\pi}{2}\right) + \frac{\pi}{2} = 0$$

$$\Rightarrow 12\alpha - \pi = 0$$

$$\alpha = \frac{\pi}{12}$$

85. Let S be the set of all solutions of the equation $\cos^{-1}(2x) - 2\cos^{-1}(\sqrt{1-x^2}) = \pi$, $x \in \left[-\frac{1}{2}, \frac{1}{2}\right]$. Then

$\sum_{x \in S} 2 \sin^{-1}(x^2 - 1)$ is equal to :

माना समीकरण $\cos^{-1}(2x) - 2 \cos^{-1}(\sqrt{1-x^2}) = \pi$, $x \in \left[-\frac{1}{2}, \frac{1}{2}\right]$ के सभी हलों का समुच्चय S है। तब $\sum_{x \in S} 2 \sin^{-1}(x^2 - 1)$ का मान है :

- (A) 0 (B) $\pi - \sin^{-1}\left(\frac{\sqrt{3}}{4}\right)$ (C) $\frac{-2\pi}{3}$ (D) $\pi - 2 \sin^{-1}\left(\frac{\sqrt{3}}{4}\right)$

Ans. C

Sol. $\cos^{-1}(2x) = \pi + 2 \cos^{-1}(\sqrt{1-x^2})$

LHS = $[0, \pi]$

For equation to be meaningful

$$\cos^{-1} 2x = \pi \text{ and } \cos^{-1}(\sqrt{1-x^2}) = 0$$

$$x = \frac{-1}{2} \text{ and } x = 0$$

which is not possible

$$\therefore x \in \phi$$

Now $\Sigma(x) = 0$

$\therefore$ Sum over empty set is always 0

86. If $y = \cos\left(\frac{\pi}{3} + \cos^{-1} \frac{x}{2}\right)$, then $(x-y)^2 + 3y^2$ is equal to _____.

यदि $y = \cos\left(\frac{\pi}{3} + \cos^{-1} \frac{x}{2}\right)$ है, तो $(x-y)^2 + 3y^2$ बराबर है _____.

Ans. 3

Sol. $y = \cos\left(\frac{\pi}{3} + \cos^{-1} \frac{x}{2}\right)$

$$= \cos\left(\frac{\pi}{3}\right) \cos\left(\cos^{-1}\left(\frac{x}{2}\right)\right) - \sin\left(\frac{\pi}{3}\right) \sin\left(\cos^{-1}\left(\frac{x}{2}\right)\right)$$

$$= \frac{1}{2} \cdot \frac{x}{2} - \frac{\sqrt{3}}{2} \cdot \sqrt{1 - \frac{x^2}{4}}$$

$$\Rightarrow 4y = x - \sqrt{3} \sqrt{4 - x^2}$$

$$\Rightarrow (4y - x)^2 = 3(4 - x^2)$$

$$\Rightarrow 16y^2 + x^2 - 8xy = 12 - 3x^2$$

$$x^2 + 4y^2 - 2xy = 3$$

$$(x - y)^2 + 3y^2 = 3$$

87. Let $S = \{x : \cos^{-1} x = \pi + \sin^{-1} x + \sin^{-1}(2x + 1)\}$. Then $\sum_{x \in S} (2x - 1)^2$ is equal to _____.

माना कि $S = \{x : \cos^{-1} x = \pi + \sin^{-1} x + \sin^{-1}(2x + 1)\}$ है, तब $\sum_{x \in S} (2x - 1)^2$ बराबर है। _____.

Ans. 5

Sol. $\cos^{-1} x = \pi + \sin^{-1} x + \sin^{-1}(2x + 1)$

$$-\frac{\pi}{2} = 2 \sin^{-1} x + \sin^{-1}(2x + 1)$$

$$-\frac{\pi}{2} - \sin^{-1} x = \sin^{-1} x + \sin^{-1}(2x + 1)$$

$$-\sin\left(\frac{\pi}{2} + \sin^{-1} x\right) = \sin(\sin^{-1} x + \sin^{-1}(2x + 1))$$

$$-\cos(\sin^{-1} x) = x \cos(\sin^{-1}(2x + 1)) + \cos(\sin^{-1} x)(2x + 1)$$

$$-\sqrt{1 - x^2} = x\sqrt{1 - (2x + 1)^2} + (2x + 1)\sqrt{1 - x^2}$$

$$x^3 - 2x - 1 = 0$$

$$(x + 1)(x^2 - x - 1) = 0$$

$$x = -1 \text{ (false)} \quad \text{or}$$

$$x^2 - x - 1 = 0$$

$$x = \frac{1 \pm \sqrt{1 + 4}}{2}$$

$$x = \frac{1 \pm \sqrt{5}}{2}$$

$$2x - 1 = \pm\sqrt{5}$$

$$2x - 1 = \sqrt{5}$$

$$\sum_{x \in S} (2x - 1)^2 = 5$$

88. Let the inverse trigonometric functions take principal values. The number of real solutions of the equation

$$2 \sin^{-1} x + 3 \cos^{-1} x = \frac{2\pi}{5} \text{ is } \underline{\hspace{2cm}}.$$

माना प्रतिलोम त्रिकोणमितीय फलन मुख्य मान लेते हैं। तो समीकरण $2 \sin^{-1} x + 3 \cos^{-1} x = \frac{2\pi}{5}$ के वास्तविक हलों की संख्या है

_____.

Ans. 0

Sol. $2 \sin^{-1} x + 2 \cos^{-1} x + \cos^{-1} x = \frac{2\pi}{5}$

$$2 \left(\frac{\pi}{2} \right) + \cos^{-1} x = \frac{2\pi}{5}$$

$$\cos^{-1} x = \frac{2\pi}{5} - \pi$$

$$\cos^{-1} x = \frac{-3\pi}{5}$$

But $\cos^{-1} x \in [0, \pi]$

$\therefore$ No real value of x

No solution

89. If $S = \left\{ x \in \mathbb{R} : \sin^{-1} \left(\frac{x+1}{\sqrt{x^2+2x+2}} \right) - \sin^{-1} \left(\frac{x}{\sqrt{x^2+1}} \right) = \frac{\pi}{4} \right\}$, then

$\sum_{x \in S} \left(\sin \left((x^2+x+5) \frac{\pi}{2} \right) - \cos \left((x^2+x+5) \pi \right) \right)$ is equal to _____.

यदि $S = \left\{ x \in \mathbb{R} : \sin^{-1} \left(\frac{x+1}{\sqrt{x^2+2x+2}} \right) - \sin^{-1} \left(\frac{x}{\sqrt{x^2+1}} \right) = \frac{\pi}{4} \right\}$, है, तो

$\sum_{x \in S} \left(\sin \left((x^2+x+5) \frac{\pi}{2} \right) - \cos \left((x^2+x+5) \pi \right) \right)$ बराबर है _____.

Ans. 4

Sol. $\sin^{-1} \left(\frac{x+1}{\sqrt{x^2+2x+2}} \right) - \sin^{-1} \left(\frac{x}{\sqrt{x^2+1}} \right) = \frac{\pi}{4}$

$$\tan^{-1} (x+1) - \tan^{-1} x = \frac{\pi}{4} \Rightarrow \tan^{-1} \left(\frac{(x+1)-x}{1+(x+1)x} \right) = \frac{\pi}{4}$$

$$\tan^{-1} \left(\frac{1}{1+x^2+x} \right) = \frac{\pi}{4} \Rightarrow \frac{1}{1+x^2+x} = 1$$

$$x^2 + x + 1 = 1 \Rightarrow x^2 + x = 0$$

$$\Rightarrow x = 0, -1 \Rightarrow S = \{0, -1\}$$

$$\begin{aligned} \text{Now, } \sum_{x \in S} \left(\sin \left((x^2 + x + 5) \right) \frac{\pi}{2} \right) - \cos \left((x^2 + x + 5) \pi \right) \\ = \left(\sin \frac{5\pi}{2} - \cos 5\pi \right) + \left(\sin \frac{5\pi}{2} - \cos 5\pi \right) = (1+1) + (1+1) = 4 \end{aligned}$$

Limit

90. For $\alpha, \beta, \gamma \in \mathbf{R}$, If $\lim_{x \rightarrow 0} \frac{x^2 \sin \alpha x + (\gamma - 1)e^{x^2}}{\sin 2x - \beta x} = 3$, then $\beta + \gamma - \alpha$ is equal to :

यदि $\alpha, \beta, \gamma \in \mathbf{R}$, के लिए $\lim_{x \rightarrow 0} \frac{x^2 \sin \alpha x + (\gamma - 1)e^{x^2}}{\sin 2x - \beta x} = 3$ है, तो $\beta + \gamma - \alpha$ बराबर है :

- (A) 7 (B) 6 (C) 4 (D) -1

Ans. A

Sol.
$$\lim_{x \rightarrow 0} \frac{x^2 (\alpha x) + (\gamma - 1) \left(1 + \frac{x^2}{1} \right)}{2x - \frac{8x^3}{6} - \beta x} = 3$$

$$\lim_{x \rightarrow 0} \frac{(\gamma - 1) + (\gamma - 1)x^2 + \alpha x^3}{(2 - \beta)x - \frac{4}{3}x^3}$$

$$\gamma - 1, \beta = 2, -\frac{3\alpha}{4} = +3 \quad \Rightarrow \alpha = 4$$

$$\beta + \gamma - \alpha = 7$$

91. If $\lim_{x \rightarrow 0} \frac{\cos(2x) + a \cos(4x) - b}{x^4}$ is finite, then $(a + b)$ is equal to :

यदि $\lim_{x \rightarrow 0} \frac{\cos(2x) + a \cos(4x) - b}{x^4}$ परिमित है, तो $(a + b)$ बराबर है :

- (A) $\frac{3}{4}$ (B) 0 (C) $\frac{1}{2}$ (D) -1

Ans. C

Sol.
$$\lim_{x \rightarrow 0} \frac{\cos(2x) + a \cos(4x) - b}{x^4}$$

$$\lim_{x \rightarrow 0} \frac{\left(1 - \frac{(2x)^2}{2!} + \frac{(2x)^4}{4!} + \dots\right) + a \left(1 - \frac{(4x)^2}{2!} + \frac{(4x)^4}{4!} - \dots\right) - b}{x^4}$$

For limit to be finite

$$1 + a - b = 0$$

$$\text{And } -2 - 8a = 0$$

$$\Rightarrow 8a = -2$$

$$\Rightarrow a = -\frac{1}{4}$$

$$\Rightarrow b = 1 + a$$

$$= 1 - \frac{1}{4}$$

$$b = -\frac{3}{4}$$

$$a + b = -\frac{1}{4} + \frac{3}{4} = \frac{1}{2}$$

92. If $\lim_{x \rightarrow 1^+} \frac{(x-1)(6 + \lambda \cos(x-1)) + \mu \sin(1-x)}{(x-1)^3} = -1$, where $\lambda, \mu \in \mathbb{R}$, then $\lambda + \mu$ is equal to :

यदि $\lim_{x \rightarrow 1^+} \frac{(x-1)(6 + \lambda \cos(x-1)) + \mu \sin(1-x)}{(x-1)^3} = -1$, $\lambda, \mu \in \mathbb{R}$ हैं, तो $\lambda + \mu$ बराबर है :

(A) 19

(B) 17

(C) 18

(D) 20

Ans. C

Sol. $\lim_{x \rightarrow 1^+} \frac{(x-1)(6 + \lambda \cos(x-1)) + \mu \sin(1-x)}{(x-1)^3} = -1$

$$\lim_{h \rightarrow 0} \frac{h(6 + \lambda \cosh) - \mu \sinh}{h^3} = -1$$

$$\lim_{h \rightarrow 0} \frac{h \left(6 + \lambda \left(1 - \frac{h^2}{2} \right) \right) - \mu \left(h - \frac{h^3}{6} \right)}{h^3}$$

$$\lim_{h \rightarrow 0} \frac{6h + \lambda h - \frac{\lambda h^3}{2} - \mu h + \frac{\mu h^3}{6}}{h^3}$$

$$\lim_{h \rightarrow 0} \frac{h(6 + \lambda - \mu) + h^3 \left(\frac{\mu}{6} - \frac{\lambda}{2} \right)}{h^3}$$

$$\lambda - \mu + 6 = 0$$

$$\dots(1)$$

&

$$\frac{\mu}{6} - \frac{\lambda}{2} = -1 \quad \dots(2)$$

$$\mu - \lambda = 6$$

$$\frac{\mu - 3\lambda}{6} = -1$$

$$\mu - 3\lambda = -6$$

$$\mu - 3\lambda = -6$$

$$2\lambda = 12$$

$$\lambda = 6$$

$$\mu = 12$$

$$\Rightarrow \lambda + \mu = 18$$

93. $\lim_{x \rightarrow 0^+} \frac{\tan \left(5(x)^{\frac{1}{3}} \right) \log_e (1 + 3x^2)}{\left(\tan^{-1} 3\sqrt{x} \right)^2 \left(e^{5(x)^{\frac{4}{3}}} - 1 \right)}$ is equal to :

$$\lim_{x \rightarrow 0^+} \frac{\tan \left(5(x)^{\frac{1}{3}} \right) \log_e (1 + 3x^2)}{\left(\tan^{-1} 3\sqrt{x} \right)^2 \left(e^{5(x)^{\frac{4}{3}}} - 1 \right)} \text{ बराबर है :}$$

(A) $\frac{1}{15}$

(B) $\frac{5}{3}$

(C) 1

(D) $\frac{1}{3}$

Ans. D

Sol.
$$\lim_{x \rightarrow 0^+} \left(\frac{\tan \left(5x^{\frac{1}{3}} \right)}{5x^{\frac{1}{3}}} \right) \cdot \left(\frac{(3\sqrt{x})^2}{\left(\tan^{-1} 3\sqrt{x} \right)^2} \right) \left(\frac{\ell(1 + 3x^2)}{3x^2} \right) \left(\frac{5x^{\frac{4}{3}}}{e^{5x^{\frac{4}{3}}} - 1} \right) \times \frac{5x^{\frac{1}{3}} \cdot 3x^2}{5x^{\frac{4}{3}} \cdot 9x}$$

$$= \frac{1}{3}$$

94. Given below are two statements :

Statement I : $\lim_{x \rightarrow 0} \left(\frac{\tan^{-1} x + \log_e \sqrt{\frac{1+x}{1-x}} - 2x}{x^5} \right) = \frac{2}{5}$

Statement II : $\lim_{x \rightarrow 1} \left(x^{\frac{2}{1-x}} \right) = \frac{1}{e^2}$

In the light of the above statements, choose the **correct** answer from the options given below.

- (A) Statement I is true but Statement II is false
- (B) Both Statement I and Statement II are true
- (C) Both Statement I and Statement II are false
- (D) Statement I is false but Statement II is true

नीचे दो कथन दिए गए हैं :

कथन I : $\lim_{x \rightarrow 0} \left(\frac{\tan^{-1} x + \log_e \sqrt{\frac{1+x}{1-x}} - 2x}{x^5} \right) = \frac{2}{5}$

कथन II : $\lim_{x \rightarrow 1} \left(x^{\frac{2}{1-x}} \right) = \frac{1}{e^2}$

इन कथनों के लिए नीचे दिए गए विकल्पों में से सही उत्तर चुनिए :

- (A) कथन I सत्य है किंतु कथन II असत्य है
- (B) कथन I असत्य है किंतु कथन II सत्य है
- (C) कथन I और कथन II दोनों सत्य हैं
- (D) कथन I और कथन II दोनों असत्य हैं

Ans. B

Sol. Statement-I : $\lim_{x \rightarrow 0} \left(\frac{\tan^{-1} x + \log_e \sqrt{\frac{1+x}{1-x}} - 2x}{x^5} \right) = \frac{2}{5}$

$$\Rightarrow \frac{\lim_{x \rightarrow 0} \left(x - \frac{x^3}{3} + \frac{x^5}{5} + \dots \right) + \frac{1}{2} \left(\left(x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} \right) + \frac{x^5}{5} + \dots \right) - \left(-x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \frac{x^5}{5} + \dots \right) - 2x}{x^5}$$

$$\lim_{x \rightarrow 0} \frac{2x^5}{x^5} + \dots = \frac{2}{5}$$

Statement-II : $\lim_{x \rightarrow 1} \left(x^{\frac{2}{1-x}} \right) = e^{\lim_{x \rightarrow 1} (x-1) \cdot \frac{2}{1-x}} = e^{-2} = \frac{1}{e^2}$

95. If $\lim_{x \rightarrow \infty} \left(\left(\frac{e}{1-e} \right) \left(\frac{1}{e} - \frac{x}{1+x} \right) \right)^x = \alpha$, then the value of $\frac{\log_e \alpha}{1 + \log_e \alpha}$ equal to :

यदि $\lim_{x \rightarrow \infty} \left(\left(\frac{e}{1-e} \right) \left(\frac{1}{e} - \frac{x}{1+x} \right) \right)^x = \alpha$ है, तो $\frac{\log_e \alpha}{1 + \log_e \alpha}$ का मान बराबर है :

- (A) e (B) e^{-2} (C) e^2 (D) e^{-1}

Ans. A

Sol. $\lim_{x \rightarrow \infty} \left(\left(\frac{e}{1-e} \right) \left(\frac{1}{e} - \frac{x}{1+x} \right) \right)^x = \alpha$

$$e^{\lim_{x \rightarrow \infty} x \left\{ \left(\frac{e}{1-e} \right) \left(\frac{1}{e} - \frac{x}{1+x} \right) - 1 \right\}} = \alpha$$

$$e^{\lim_{x \rightarrow \infty} x \left\{ \frac{e(1+x-ex)}{(1-e)e(1+x)} - 1 \right\}} = \alpha$$

$$e^{\lim_{x \rightarrow \infty} x \left\{ \frac{e+ex-e^2x-e+e^2-ex+e^2x}{e+e^2+ex-e^2x} \right\}} = \alpha$$

$$e^{\lim_{x \rightarrow \infty} \frac{x e^2}{e - e^2 + x(e - e^2)}} = \alpha$$

$$e^{\frac{e^2}{e - e^2}} = \alpha \quad \Rightarrow \alpha = e^{\frac{e}{1-e}}$$

$$\text{Now } \frac{\log \alpha}{1 + \log_e \alpha} = \frac{\log_e \left(e^{\frac{e}{1-e}} \right)}{1 + \log_e \left(e^{\frac{e}{1-e}} \right)} = \frac{\frac{e}{1-e}}{1 + \frac{e}{1-e}} \Rightarrow e$$

96. Let $f: \mathbb{R} - \{0\} \rightarrow \mathbb{R}$ be a function such that $f(x) - 6f\left(\frac{1}{x}\right) = \frac{35}{3x} - \frac{5}{2}$. If the $\lim_{x \rightarrow 0} \left(\frac{1}{\alpha x} + f(x) \right) = \beta$; $\alpha, \beta \in \mathbb{R}$, then

$\alpha + 2\beta$ is equal to :

माना कि $f: \mathbb{R} - \{0\} \rightarrow \mathbb{R}$ एक फलन इस प्रकार है कि $f(x) - 6f\left(\frac{1}{x}\right) = \frac{35}{3x} - \frac{5}{2}$ है। यदि $\lim_{x \rightarrow 0} \left(\frac{1}{\alpha x} + f(x) \right) = \beta$; $\alpha, \beta \in \mathbb{R}$ है, तो $\alpha + 2\beta$ बराबर है :

(A) 3

(B) 5

(C) 4

(D) 6

Ans. C

Sol. $f(x) - 6f\left(\frac{1}{x}\right) = \frac{35}{3x} - \frac{5}{2} \quad \dots(1)$

Replace $x \rightarrow \frac{1}{x}$

$$f\left(\frac{1}{x}\right) - 6f(x) = \frac{35x}{3} - \frac{5}{2} \quad \dots(2)$$

from (1) and (2)

$$f(x) = -2x - \frac{1}{3x} + \frac{1}{2}$$

Now,

$$\beta = \lim_{x \rightarrow 0} \left(\frac{1}{\alpha x} - 2x - \frac{1}{3x} + \frac{1}{2} \right)$$

$$\Rightarrow \alpha = 3, \beta = \frac{1}{2}$$

$$\text{So, } \alpha + 2\beta = 3 + 1 = 4$$

97. For some a, b , let $f(x) = \begin{vmatrix} a + \frac{\sin x}{x} & 1 & b \\ a & 1 + \frac{\sin x}{x} & b \\ a & 1 & b + \frac{\sin x}{x} \end{vmatrix}$, $x \neq 0$, $\lim_{x \rightarrow 0} f(x) = \lambda + \mu a + \nu b$. Then

$(\lambda + \mu + \nu)^2$ is equal to :

किसी a, b , के लिए माना कि $f(x) = \begin{vmatrix} a + \frac{\sin x}{x} & 1 & b \\ a & 1 + \frac{\sin x}{x} & b \\ a & 1 & b + \frac{\sin x}{x} \end{vmatrix}$, $x \neq 0$, $\lim_{x \rightarrow 0} f(x) = \lambda + \mu a + \nu b$ है। तब

$(\lambda + \mu + \nu)^2$ बराबर है :

(A) 9

(B) 36

(C) 16

(D) 25

Ans. C

Sol. $\lim_{x \rightarrow 0} f(x) = \begin{vmatrix} a+1 & 1 & b \\ a & 2 & b \\ a & 1 & b+1 \end{vmatrix}$

$$= (a+1)(b+2) - a(1) + a(-b)$$

$$= 2 + a + b$$

$$\lambda = 2, \mu = 1, \nu = 1$$

$$(\lambda + \mu + \nu)^2 = 16$$

98. $\lim_{n \rightarrow \infty} \frac{(1^2 - 1)(n-1) + (2^2 - 2)(n-2) + \dots + ((n-1)^2 - (n-1)) \cdot 1}{(1^3 + 2^3 + \dots + n^3) - (1^2 + 2^2 + \dots + n^2)}$ is equal to :

$\lim_{n \rightarrow \infty} \frac{(1^2 - 1)(n-1) + (2^2 - 2)(n-2) + \dots + ((n-1)^2 - (n-1)) \cdot 1}{(1^3 + 2^3 + \dots + n^3) - (1^2 + 2^2 + \dots + n^2)}$ बराबर है :

(A) $\frac{2}{3}$

(B) $\frac{3}{4}$

(C) $\frac{1}{2}$

(D) $\frac{1}{3}$

Ans. D

Sol. $\lim_{n \rightarrow \infty} \frac{(1^2 - 1)(n-1) + (2^2 - 2)(n-2) + \dots + ((n-1)^2 - (n-1)) \times 1}{(1^3 + 2^3 + \dots + n^3)(1^2 + 2^2 + \dots + n^2)}$

$$\text{Numerator} = \sum_{r=1}^{n-1} ((r-1)^2 - (r-1))(n-r)$$

$$= \sum_{r=1}^{n-1} (r-1) - (r-2)(n-r)$$

$$= \sum_{r=1}^{n-1} -r^3 + (n+3)r^2 - (2+3n)r + 2n$$

We will take term with the greatest power of n

$$= \frac{-1}{4}n^4 + \frac{1}{3} + \frac{1}{3}n^4 = \frac{1}{12}n^4$$

$$\text{Denominator} = \sum_{r=1}^n r^3 - \sum_{r=1}^n r^2$$

$$= \left(\frac{n(n+1)^2}{2} \right) - \left(\frac{n(n+1)(2n+1)}{6} \right)$$

Greatest power of n is $\frac{n^4}{4}$

$$\lim_{n \rightarrow \infty} \frac{\frac{1}{12}n^4}{\frac{n^4}{4}} = \frac{1}{3}$$

99. Let a circle passing through (2, 0) have its centre at the point (h, k). Let (x_c, y_c) be the point of intersection of the lines $3x + 5y = 1$ and $(2+c)x + 5c^2y = 1$. If $h = \lim_{c \rightarrow 1} x_c$ and $k = \lim_{c \rightarrow 1} y_c$, then the equation of the circle is:

माना (2, 0) से होकर जाने वाले एक वृत्त का केन्द्र (h, k) है। माना रेखाओं $3x + 5y = 1$ तथा $(2+c)x + 5c^2y = 1$ का प्रतिच्छेदन बिन्दु (x_c, y_c) है। यदि $h = \lim_{c \rightarrow 1} x_c$ तथा $k = \lim_{c \rightarrow 1} y_c$ है, तो वृत्त का समीकरण है :

- (A) $25x^2 + 25y^2 - 2x + 2y - 60 = 0$ (B) $5x^2 + 5y^2 - 4x + 2y - 12 = 0$
(C) $25x^2 + 25y^2 - 20x + 2y - 60 = 0$ (D) $5x^2 + 5y^2 - 4x - 2y - 12 = 0$

Ans. C

Sol. $3x + 5y = 1$

$$(2+c)x + 2c^2y = 1$$

$$(3c^2 - c - 2)x = (c^2 - 1)$$

$$x = \frac{c^2 - 1}{3c^2 - c - 2} \text{ and } y = \frac{1}{5} \left(1 - \frac{3(c^2 - 1)}{3c^2 - c - 2} \right) = \frac{1 - c}{5(3c^2 - c - 2)}$$

$$h = \lim_{c \rightarrow 1} \frac{(1-c)}{(1-c)} \left(\frac{1+c}{2+3c} \right) = \frac{2}{5}$$

$$k = \lim_{c \rightarrow 1} \frac{1-c}{5(c-1)(3c+2)} = \lim_{c \rightarrow 1} \frac{-1}{5(3c+12)} = -\frac{1}{25}$$

$$\text{Centre of circle} = \left(\frac{2}{5}, -\frac{1}{25} \right)$$

$$\text{Now radius of circle} = \sqrt{\left(2 - \frac{2}{5} \right)^2 + \left(\frac{1}{25} \right)^2} = \sqrt{\frac{64}{25} + \frac{1}{625}}$$

Equation of circle is

$$\left(x - \frac{2}{5} \right)^2 + \left(y + \frac{1}{25} \right)^2 = \frac{64}{25} + \frac{1}{625}$$

$$x^2 + y^2 - \frac{4}{5}x + \frac{2}{25}y + \frac{4}{25} + \frac{1}{625} = \frac{64}{25} + \frac{1}{625}$$

$$25x^2 + 25y^2 - 20x + 2y + 4 = 64$$

$$25x^2 + 25y^2 - 20x + 2y - 60 = 0$$

100. $\lim_{x \rightarrow 0} \frac{e - (1+2x)^{\frac{1}{2x}}}{x}$ is equal to :

$$\lim_{x \rightarrow 0} \frac{e - (1+2x)^{\frac{1}{2x}}}{x} \text{ बराबर है :}$$

(A) e (B) $\frac{-2}{e}$ (C) 0 (D) $e - e^2$

Ans. A

Sol. $\lim_{x \rightarrow 0} \frac{e - e^{\frac{1}{2x} \ln(1+2x)}}{x}$

$$= \lim_{x \rightarrow 0} (-e) \frac{\left(e^{\frac{\ln(1+2x)}{2x} - 1} - 1 \right)}{x}$$

$$= \lim_{x \rightarrow 0} (-e) \frac{\ln(1+2x) - 2x}{2x^2}$$

$$= (-e) \times (-1) \frac{4}{2 \times 2} = e$$